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MEMORANDUM TO: Mr. J. C. Logue

SUBJECT: 604X Design

The following are elements of the design of the 604X as I see them. Major credit of the design should go to Dr. E. Amdahl.

I. Design Assumptions

A. 4 instructions per microsecond or faster using one CPU.
B. NPL upwards compatible in instruction set and formats.
C. 1μs memory. Interleaving up to 16-fold.
D. ACP circuitry at 2-4 nanoseconds/stage, or cycle time from 30-60 nanoseconds.
E. Second CPU available.
F. Addressed to the superlarge machine users (possibly including the 7950 customer with some changes in instruction set).

II. Instruction Set

A. Upwards compatible with NPL.
B. To utilize the multi-accumulator design the following are added:

1. RRX instructions which allow a 24 bit memory operand address and a second R field, such that memory operand can interact with one accumulator and the result can go to a different accumulator. (This effectively removes the chief complaint against NPL instruction design.)

2. Fixed-point type instructions in "floating - point" registers. This is critical in the high-performance machines, since the I-box accumulators have special interlocking constraints because they double as index (and base) registers.

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3. Special features to allow the testing of E-box arithmetic results, such that the results of the test can become the basis of a future branch instruction. This requires corresponding adjustment in branch type instructions.

C. Instructions efficient in lower machines may become extremely slow in 604 X. Their existence should not be construed as invitation for usage. This has far-reaching implications of course, and to do justice to the design a new programming system (stressing scientific usage) is a must.

III. Hardware

I/O same as 604

I-box for the most part similar to 604, with changes in

(a) E-box instruction handling
(b) conditional branch handling
(c) address handling
(d) I-fetch operand handling

E-box completely new, with emphasis on performance:

static E-box oriented lookahead (8-16 levels)
microcoded instructions in lookahead.

16 (or 8) accumulators in E-box.
"push-through" arithmetic unit design.

Memory: 16-interleaved 1μs storage (128-256 K 64 bit words)
1 or 2 bulk storages (512K words each)

IV. Store-fetch Interlock

All E-box stores are preceded by a destructive fetch, leaving a bit-configuration clearly recognizable as invalid data.

All E-box stores terminate with the arrival of valid data in the store address.

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Because of this interlock, all fetches proceed normally without testing for conflicts (as was the case with the 7030). Those fetches which conflict with "pending stores" are caught automatically by the invalid data condition. The fetch can then either (a) wait until the store action is complete, or (b) initiate a search in the E-box to locate the store operand and perform "forwarding."

This interlock makes it possible for the lookahead to be normally disassociated from the I-box, and a strong coupling occurs only infrequently.

(The scheme is harder to implement for partial stores. I recommend that half-word I-box arithmetic not be installed for the E-box)

V. Conditional Branches

Unconditional branches are handled normally, as are conditional branches based on I-box data.

Conditional branches based on pre-set conditions require minimum coupling between E-box and I-box (communication of several bits) and is not a serious problem.

Conditional branches based on conditions unknown to I-box during decoding time (the E-operation to be tested has not yet been performed) has been a headache on the 7030, now solved in the following manner:

a. Both branches are followed (the "probable" branch being the next instruction in sequence and is handled normally) and one instruction is fetched from the improbable branch.

b. If the instruction is a fetch, the fetch operand is also obtained.

c. Since the instructions and the operands may refer to the I-box, it is advantageous to put them in the I-box.

d. When the true conditions are known, one of the paths is cancelled by the I-box.

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VI. Lookahead

The lookahead has an instruction stack (INS) and a memory operand stack (MOS). If their sizes are the same, some simplification in design can be achieved; otherwise each of the instruction levels would have to carry one more address (memory operand stack address).

The operation is dependent on the I-box and E-box. Only data relevant to E-box instructions may enter the lookahead.

Whenever the I-box decodes an E-box instruction, the INS is loaded with microcoded instructions to simplify E-box sequencing. If the E-instruction is RR-type, little more needs to be done. If RX fetch type, I-box initiates a fetch returnable to the MOS, and disguises the instruction as a microcoded R$\tilde{R}$-type ($\tilde{R}$ refers to the MOS). If RX store type, I-box initiates destructive fetch (for interlock), and puts address in MOS, again using an R$\tilde{R}$-type microcode. All R$\tilde{R}$-code returnable to MOS implies a store operation. The store address should best be next to the store operand address. RRX-type instructions are handled similarly.

RX-load and RX-load negative instructions can be accelerated by bypassing the lookahead stage. Details are to be worked out later.

All incomplete fetches lead to "operand not yet available" tag on the MOS level. The tag is removed on operand arrival, before this, the level contents cannot be used.

VII. E-Sequencing

The arithmetic unit is designed to allow several operations (about 4 maximum) to occur at once. This means several instructions, of course. Whenever facilities are available, operands are pushed through them. To do this effectively the operation code must be predigested into microcode, and operands accompanied by microcode move through the hardware not unlike a document with a routing slip.

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Whenever an operand is destined for a register, the register is made to carry an "operand not yet available" tag until the routing is complete. This interlock is sufficient to guarantee logical coherence for a simple "push-through" E-box design. With a little extra hardware it is possible to perform operations using the result of the previous operation(s) (simple forwarding, just name the internal registers of the arithmetic unit and include them in the routing). With more extra hardware it is possible to perform operations out of sequence. The limit is clearly 1 instruction per machine cycle.

VIII. Other CPU's

It is my opinion that a second CPU for the 604X is an asset; it is much more difficult to justify additional CPUs beyond this.

A 16-fold interleaved memory can support two CPUs without drastic loss of performance, and one obtains the benefit of reliability, performance gain with relatively small extra investment. The intercommunication required is a minimum, and a good monitor program can be designed for both the mono- and duo- configurations.

IX. Alternative Designs

The static lookahead design above is clean and efficient for machines with expensive local memory.

Bengt Carlson's machine concept (Sept. 1963) consists of a larger (to 64 words) INS, a larger (to 64 words) MOS, plus the ability to loop back in INS, the ability to index in MOS. He did not resolve the problem of importing information into INS-MOS, nor the exportation.

The "RB" design (Spring 1963) solved these problems by retaining the "external" lookahead stream yet allowing a true stored-program inner computer stream to regurgitate. Further, block-transmittals into and out of the inner computer are premitted. If Carlson's scheme were to be fully implemented, it may not look too different from this. A highspeed TD memory is a prerequisite.

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It is my opinion that for inner computer storage of about 100 words the external stream is needed except for very special purpose operations. At 1000 words or so, however, the situation changes, and block transmittal can replace the external lookahead mechanism.

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